

Exam: Effect Sizes & Statistical Measures

Practice Exam with Solutions | Basic Level

1. [Application] (4 Pts.) $n_1=n_2=25$, $M_1=85$, $M_2=72$, $SD_{\text{pooled}}=16$. Cohen d , interpret, Hedges g , 95% CI.
2. [Application] (4 Pts.) 2x2 table: treatment 15/35, control 25/25. OR, RR, ARR, NNT.
3. [Open question] (2 Pts.) Effect size for ANOVA with 3 groups? Thresholds?
4. [Open question] (2 Pts.) Why Hedges g instead of Cohen d for $n<20$?
5. [Open question] (3 Pts.) NNT=5. Clinical meaning?

SOLUTIONS

1. $d=0.813$ (large), Hedges $g=0.800$, $CI=[0.215;1.411]$.
2. $OR=0.429$, $RR=0.6$, $ARR=0.2$, $NNT=5$.
3. η^2 : 0.01 small, 0.06 medium, 0.14 large.
4. Cohen d overestimates population effect. Hedges g corrects by factor.
5. Treat 5 patients for one additional benefit. Lower NNT = more effective.